2nd Year 1st Semester Examination-2020

Department: *Computer Science & Engineering*

Subject: Object Oriented Programming Language — Course Code: CSE-301

Total Marks: 70 — Time: 2 (two) hours

Answer any 03 (three) questions.

01 a) Which are the best features of OOPs and why explain? 6

b) What are the advantages of OOP over structural programming? 6

c) Is encapsulation and data hiding same? 6

d) Assign and print the roll number, phone number and address of two students having names "X" and "Y" respectively by creating two objects of class "Student". 5.33

02 a) Define copy constructor. 3.33

b) Distinction between static and non-static member. 4

c) What are the uses of access modifiers in C++? Explain through an OOP program. 8

d) Can we have both default constructor and parameterized constructor in the same class? Explain through an OOP program. 8

03 a) What is an exception? Discuss how try-catch-finally blocks works in java. 4

b) What is exception handling? Using exception handling mechanism write a program to manage Arithmetic exception in java. 5

c) What happens if an exception is not caught? What will be the output of the following program : 5

```
Class A{
  Public static void main(String[] args){
   Try{
    int [] a = {5,10,15};
    System.out.println(a[5]);
    System.out.println(10/0);
    catch(ArithmeticException e){
     System.out.println("Arithmetic Exception");
    }
   catch(ArrayIndexOutOfBoundsException e){
    System.out.println("Hello world");
   }
    System.out.println("Exception Handling");
  }
}
```

d) What is inline function? Write down the advantages and limitations of inline function. 4.33

e) What is template? What would be the output of the following program : 5

```
#include<iostream>
using namespace std;
template<typename T>
void print_data(T output)
 {
   cout<<output<<endl;
 }
 int main()
```

```
{
  double d = 5.5;
  int a = 10;
  print_data(d);
  print_data(a);
  return 0;
}
```

04	a)	What is new and delete operator? What are their advantages?	4
	b)	Define multiple inheritance. Give an example of ambiguity in multiple inheritance.	4
	c)	Explain the concept of operator overloading. What are the ways to overload an operator in c++?	6
	d)	How to prevent overriding and inheritance using final keyword? Explain with example.	4
	e)	Define static keyword. Write down the differences among public, private and protected access specifier in a class.	5.33
05	a)	What is the difference between overloading and overriding concepts in C++? Explain the usage of these concepts with suitable example code in C++/Java?	8
	b)	How to create a thread? Mention the advantages of multithreading.	6
	c)	What will be the output of the following program?	5

```
abstract class Bike{
  Bike(){System.out.println("bike is created");}
  abstract void run();
  void changeGear(){System.out.println("gear changed");}
}

 class Honda extends Bike{
void run(){System.out.println("running safely..");}
}

 class TestAbstraction2{
public static void main(String args[]){
 Bike obj = new Honda();
 obj.run();
 obj.changeGear();
 }
}
```

	d)	Show the life cycle of a thread.	4.33